



Environmental Impact Analysis

The core objective of this study is to estimate the total amount of emissions produced by small vehicles in Australia. While the number of small vehicles is known, the corresponding emission levels are not directly available. To address this gap, prior knowledge derived from an emission dataset is incorporated into the analysis. The mean and standard deviation of this prior distribution are used to model and predict the total emissions generated by Australia's small-vehicle fleet. By examining historical emission patterns, the study also visualises long-term trends and highlights the potential risks associated with increasing emissions if sustainability measures are not prioritised. Additionally, the analysis considers the positive impact of electric vehicle adoption in reducing overall emissions and supporting environmentally responsible transport planning.

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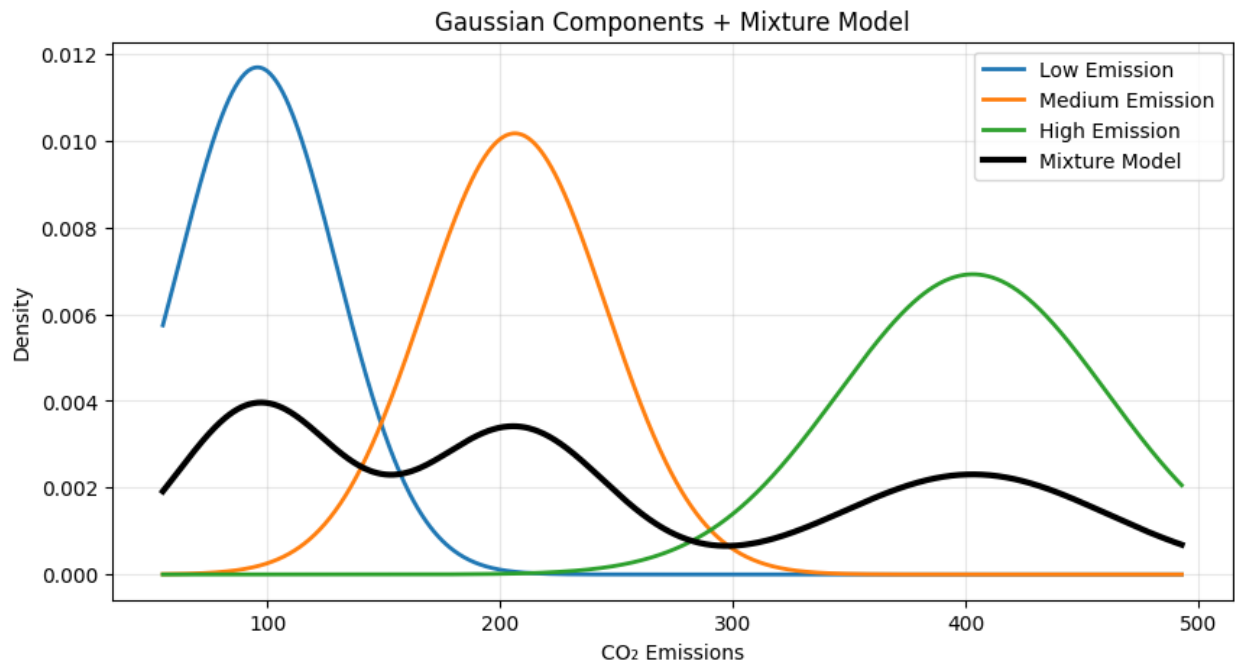
This dataset contains observations of small vehicles register from 2021 to 2026 in Australia ©[AAA Index](#). The amount of vehicle count can be dynamically integrated with the webpage through a pipeline.

- Differentiating Fuel Types. [ICE vs EV]
- Group by state.
- Aggregate total number of cars by states.
- Later find the amount of CO₂ emission (possibly green-house emission) from ICE vehicles.
- Providing a sense of sustainability towards EV users (among n%), amount of CO₂ reduced from choosing EV.

Possible Simulated Emission

- Diving into dataset attributes.
- Categorical variable analysis to get precise idea about distribution, to have more intuition about the observation of the dataset. For example, the distribution of [*'Vehicle Type', 'Fuel Type', 'Road Type', 'Traffic Conditions'*] with the categorical target variable *'Emission Level'*. Later, forming Joint Distribution.
- Continuous variable analysis. This includes generation of pair-plot to identify correlation among variables. The Emission Level was a rule-based categorization from CO₂ emission attribute values.
 - Distribution of Greenhouse gases [CO₂, Nox, PM2.5, VOC, SO₂ Emissions]
- Bayesian Analysis

- Identifying prior (had miscalculation)
- Creating Gaussian Mixture Model from three emission level categories.
- Generating Total Emission with 95% credible interval.
- “Demo” prediction of Total CO₂ Emissions by Year



- Dataset does not have **zero emission** observation (no EV's). Hence, the following dataset Environmental Protection Agency [EPA](#) (in USA) will be considered.

EPA

- Generate GMM. (one Normally Distributed, preferably two Gaussian Models)